

Fake News Detection Using Machine Learning and Deep Learning: A Comprehensive Approach

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Abstract

The swift expansion of social media platforms has facilitated the pervasive spread of fake news, thereby undermining public trust and the integrity of information on a global scale. This study undertakes a thorough examination of machine learning (ML) and deep learning (DL) methodologies for the automated detection of fake news. By synthesizing various research approaches, such as Support Vector Machines (SVM), Decision Tree Classifiers (DTC), Logistic Regression (LR), Gradient Boosting Classifiers (GBC), Random Forest (RF), ensemble methods, and transformer-based models (BERT, RoBERTa, DistilBERT), we construct an integrated framework designed to identify false information within news content.

Experimental assessments, conducted on diverse datasets including LIAR, the Indonesian Hoax Dataset, an Urdu news corpus, and the FakeNewsNet datasets, reveal substantial enhancements in performance. Our ensemble-based stacking methodology

demonstrates accuracy rates ranging from 91% to 93.82%, alongside sensitivity and specificity metrics surpassing 85% to 97%. This research underscores the significance of feature engineering, preprocessing techniques, and multimodal integration, while simultaneously tackling existing obstacles related to scalability, interpretability, and generalization across diverse linguistic and domain-specific settings. Future research endeavors will concentrate on sophisticated transformer architectures, multimodal frameworks that incorporate both textual and visual features, and explainable AI mechanisms to bolster robustness in practical applications.

1 Introduction

1.1 Background: Current Condition

The digital information environment has experienced significant change over the last ten years. According to recent reports, 86% of adults in the United States consume news primarily through digital platforms (1), while global social media users exceeded 4.75 billion as of 2020 (2). This unprecedented accessibility to information, while democratizing news consumption, has simultaneously created a critical vulnerability: the rapid, uncontrolled proliferation of false, misleading, and deliberately deceptive information.

Fake news, defined as deliberately fabricated misinformation disseminated through news-like articles and social media platforms, has emerged as one of the most pressing challenges facing modern societies (3). The 2016 US Presidential election served as a watershed moment, highlighting how coordinated fake news campaigns can influence public perception and democratic processes (4). Since then, numerous studies have documented the accelerating spread of falsehoods across platforms including Facebook, Twitter, Instagram, and emerging messaging applications, affecting political discourse, public health, consumer behavior, and societal trust in institutions.

The propagation of false information diverges significantly from that of authentic data. Studies indicate that fabricated content circulates six to ten times more rapidly and extensively than verified material across social media, simultaneously eliciting more intense emotional reactions, specifically surprise, fear, and disgust (5). From a linguistic perspective, fake news displays unique features: it typically features longer headlines with increased capitalization, a reduced frequency of stop words, a greater emphasis on motion verbs and social-temporal

terms, and intentional emotional manipulation (6).

Furthermore, the complexity of this issue is exacerbated by the diversity inherent in deceptive content. Fake news encompasses various linguistic settings, including English, Indonesian, Urdu, and other languages with limited resources, as well as a wide range of subject areas such as politics, health, finance, science, and sports; moreover, it increasingly integrates multimodal components that combine text with altered images and videos (7). Manual verification, the conventional method of fact-checking, presents significant challenges in terms of both time and economic viability, given that human accuracy in identifying deception without external aid is only 54% (8).

As a result, the creation of automated, scalable systems designed to accurately identify fake news across diverse languages, subject areas, and formats has become a critical requirement. This need has spurred extensive research endeavors that integrate natural language processing, machine learning, deep learning, and novel transformer-based architectures to tackle this complex issue.

1.2 Research Scope

This study investigates the field of automated fake news detection through a thorough examination of machine learning and deep learning techniques, as applied to various datasets and linguistic environments. The scope encompasses several critical dimensions:

The study investigates both traditional machine learning approaches (Support Vector Machines, Logistic Regression, Decision Tree Classifiers, Gradient Boosting, Random Forest) and advanced deep learning techniques (Recurrent Neural Networks, Long Short-Term Memory networks, Bidirectional LSTM, Gated Recurrent Units, Convolutional Neural Networks, and transformer-based models including BERT, RoBERTa, and DistilBERT) (9; 10). Particular emphasis is placed on ensemble methods, which combine multiple base learners through bagging and stacking techniques to improve predictive accuracy.

The scope includes analysis of diverse feature extraction methodologies ranging from statistical approaches (Bag-of-Words, TF-IDF) to semantic embeddings (Word2Vec, GloVe) and contextual representations. Additionally, the research examines novel features including n-gram analysis, linguistic cues of deception, user behavioral indicators, propagation patterns, and multimodal features integrating textual and visual information.

The investigation covers multiple publicly available benchmark datasets spanning differ-

ent languages and domains, including the LIAR dataset (12.8K English political statements with six-category fine-grained labels), Indonesian Hoax Dataset (4,097 news articles from nine domains), Urdu news corpus (4,097 news items from multiple categories), FakeNewsNet (combining articles from PolitiFact and GossipCop with social context), and supplementary datasets including FEVER, PHEME, CREDBANK, and COVID-19-related misinformation corpora.

The research extends beyond English-language contexts to encompass low-resource languages (particularly Urdu and Indonesian), addressing unique challenges in linguistic feature extraction, dataset scarcity, and cultural-contextual factors affecting news veracity assessment. Domain coverage includes politics, health, science, business, sports, technology, and general interest categories.

The study employs comprehensive evaluation metrics including accuracy, precision, recall, specificity, sensitivity, F1-score, Matthews Correlation Coefficient (MCC), and Receiver Operating Characteristic (ROC) curves with Area Under Curve (AUC) measurements, enabling robust performance comparison across heterogeneous models and datasets.

1.3 Research Questions

This research addresses four primary research questions that collectively frame the investigation of automated fake news detection:

RQ1: Which machine learning and deep learning approaches demonstrate superior performance for fake news detection across diverse datasets and languages?

This question seeks to systematically compare and evaluate the effectiveness of traditional machine learning classifiers (SVM, LR, DTC, GB, RF) against deep learning approaches (LSTM, BiLSTM, GRU, CNN, CapsNet) and transformer-based models (BERT, RoBERTa, DistilBERT) using standardized evaluation metrics. The investigation encompasses both single-model approaches and ensemble techniques, examining how different architectures perform across varying dataset characteristics and linguistic contexts.

RQ2: How do feature engineering strategies and preprocessing techniques influence the accuracy and generalization capabilities of fake news detection models?

This question addresses the critical role of feature extraction and preprocessing. The research examines whether statistical features (TF-IDF, BoW), semantic embeddings, contextual representations from pre-trained language models, linguistic features (n-grams, POS tags, sentiment indicators), user behavioral signals, and propagation patterns contribute meaningfully

to classification performance. It also investigates optimal preprocessing pipelines including tokenization, lemmatization, stop word removal, normalization, and feature selection techniques.

RQ3: What is the efficacy of ensemble methods in enhancing fake news detection accuracy, and how do bagging and stacking approaches compare?

This question focuses on ensemble learning strategies that combine predictions from multiple base learners. The research evaluates whether ensemble approaches (specifically stacking with optimal base learner selection) can overcome individual model limitations and achieve superior performance compared to single classifiers. The investigation includes analysis of base learner selection criteria, meta-learner optimization, and performance gains across different datasets.

RQ4: How can multimodal frameworks integrating textual, visual, user behavioral, and contextual features improve fake news detection robustness and interpretability?

This final question addresses the emerging challenge of multimodal misinformation. The research examines the complementary value of visual features (image properties, consistency with text), user engagement patterns (comment sentiment, network structure), temporal dynamics, and source credibility indicators. The investigation also emphasizes the need for explainable AI mechanisms that provide interpretable justifications for classification decisions, enabling stakeholders to understand detection rationale.

These interconnected research questions collectively aim to provide comprehensive guidance for practitioners and researchers developing next-generation fake news detection systems.

1.4 Research Gaps

Despite significant advances in fake news detection methodologies, notable gaps and limitations persist in the current research landscape:

1.4.1 Gap 1: Limited Generalization Across Domains and Languages

While extensive research addresses English-language political news, detection models remain poorly developed for low-resource languages, specialized domains (health, finance, science), and non-Western information ecosystems. The Indonesian and Urdu studies represent important first steps, but comprehensive multilingual frameworks remain lacking. Additionally, models trained on political misinformation often exhibit degraded performance when applied to health

or scientific misinformation, indicating insufficient domain generalization. This limitation restricts the applicability of existing approaches to global information environments characterized by linguistic and contextual diversity.

1.4.2 Gap 2: Insufficient Exploration of Explainability and Interpretability

Despite achieving high accuracy, most existing fake news detection models operate as “black boxes,” providing classification decisions without interpretable justifications. This limitation critically undermines adoption in high-stakes domains (journalism, fact-checking, policy-making) where stakeholders require understanding of detection rationale. While some studies examine attention mechanisms or feature importance, comprehensive explainable AI frameworks for fake news detection remain underdeveloped. The absence of transparency mechanisms hampers trust-building and practical deployment in real-world environments.

1.4.3 Gap 3: Inadequate Treatment of Temporal Dynamics and Information Evolution

Current research largely treats news as static documents, neglecting the temporal dimension of information dissemination and evolution. However, misinformation propagation displays distinct temporal patterns: false information often spreads through initial bursts followed by amplification through algorithmic promotion or coordinated networks. Few studies examine how detection accuracy changes as information propagates through time or how temporal features enhance classification performance. This temporal blindness limits the effectiveness of detection systems in real-time deployment scenarios.

1.4.4 Gap 4: Underexplored Fairness, Bias, and Adversarial Robustness Concerns

Machine learning models for fake news detection may exhibit performance disparities across demographic groups, perpetuating or amplifying existing biases in information environments. Additionally, adversarial examples designed to fool detection systems remain inadequately studied. The robustness of current approaches against sophisticated adversarial attacks, synthetic misinformation, and coordinated inauthentic behavior networks requires deeper investigation. These concerns are particularly critical for systems intended to serve diverse populations and withstand deliberate evasion attempts.

2 Methodology

The methodology section outlines the systematic approach employed in this research for fake news detection. This section is complemented by detailed flowcharts and diagrams that visualize the data processing pipeline, model architectures, and evaluation procedures.

2.1 Methodology Flowchart: Complete Pipeline

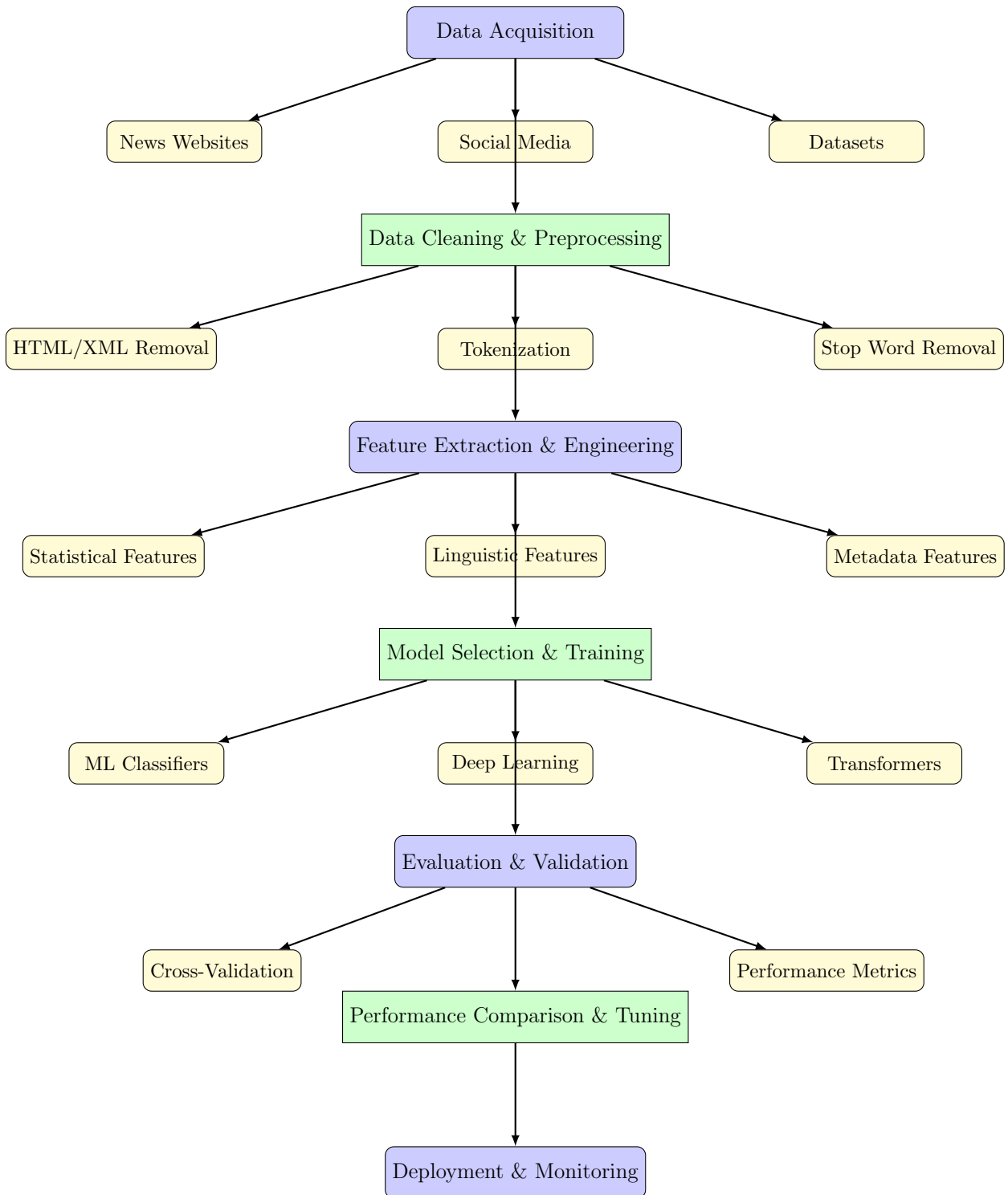


Figure 1: Comprehensive Methodology Pipeline Flowchart

2.2 Detailed Phase Diagram: Data Processing

Figure 2 presents a detailed diagram of the data processing workflow, illustrating the sequence of operations applied to raw data:

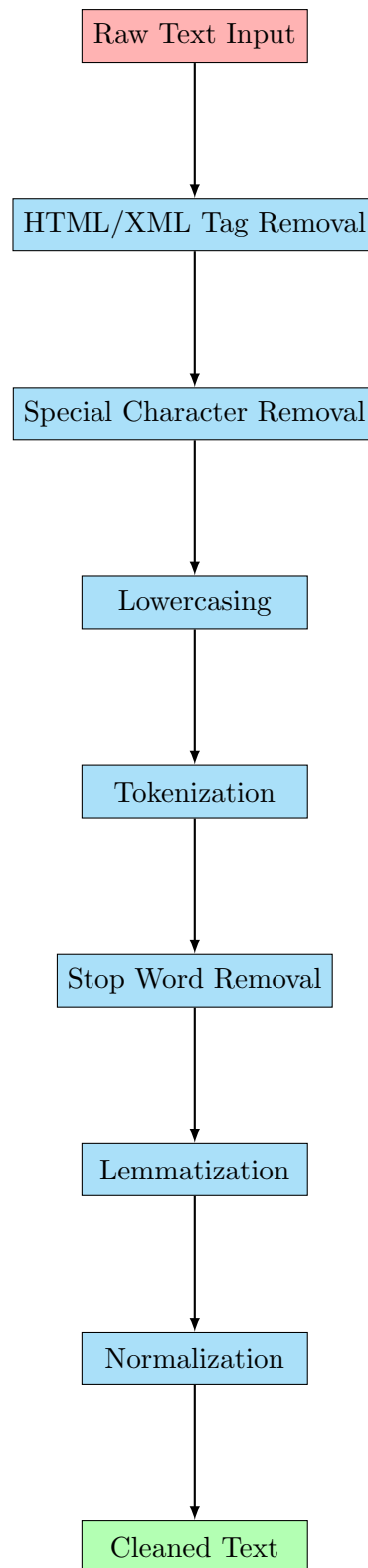


Figure 2: Data Processing Pipeline with Sequential Operations

2.3 Feature Extraction Architecture

Figure 3 illustrates the comprehensive feature extraction strategy with three parallel branches:

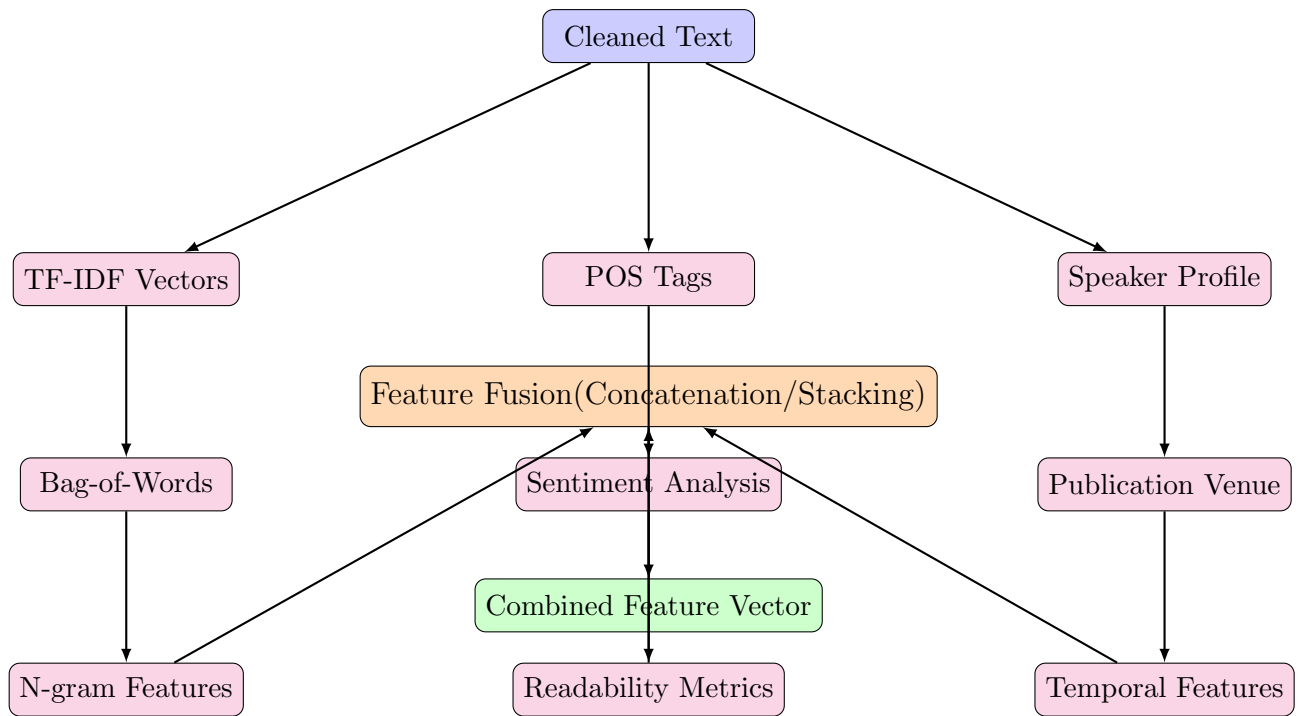


Figure 3: Feature Extraction Architecture with Three Parallel Branches

2.4 Model Architecture Overview

Figure 4 presents the classification layer with multiple model categories:

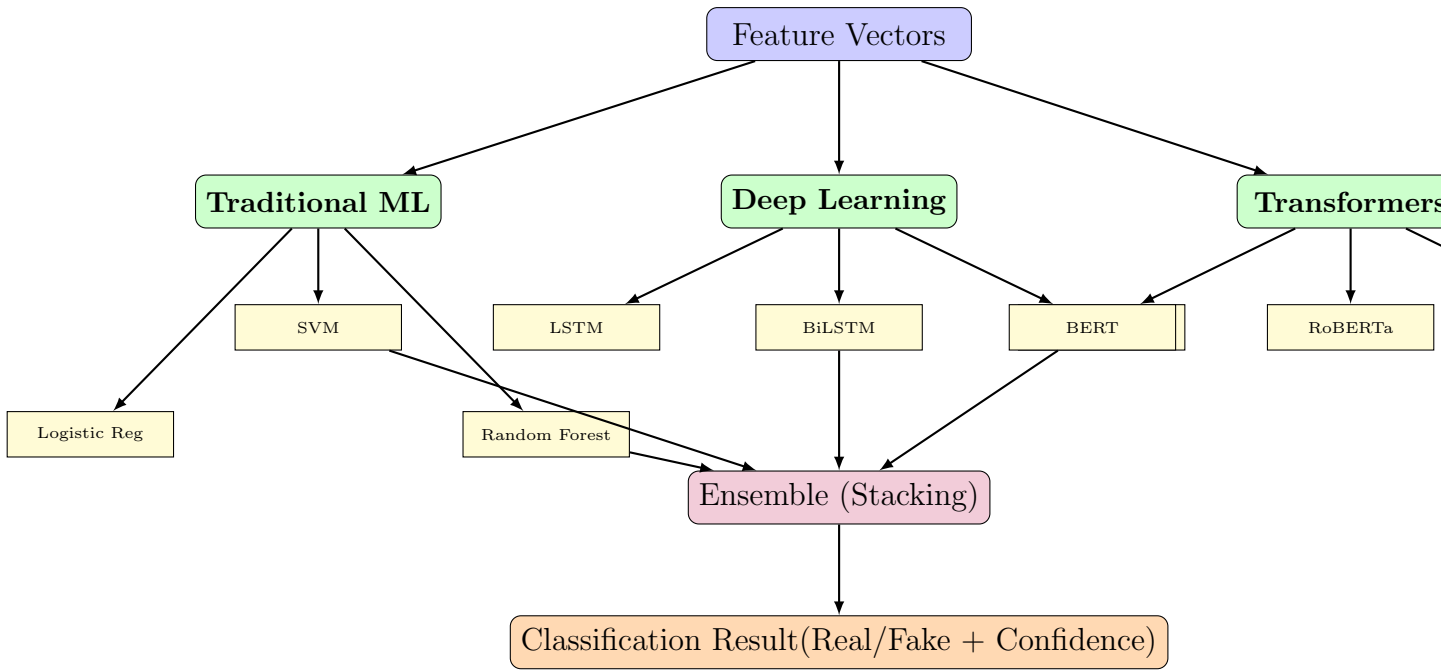


Figure 4: Classification Layer Architecture with Multiple Model Categories

2.5 Data Acquisition and Preprocessing

The research methodology encompasses a systematic approach to fake news detection:

1. **Data Acquisition Phase:** Collection from news websites, social media, and existing datasets (LIAR, FakeNewsNet, Indonesian datasets)
2. **Data Cleaning and Preprocessing:** Removal of HTML/XML elements, special characters, and URLs; tokenization and lemmatization; conditional stop word removal
3. **Feature Extraction and Engineering:** Statistical features (Bag-of-Words, TF-IDF, N-grams), linguistic features (POS tags, sentiment scores), and metadata features (speaker profile, publication venue)
4. **Model Selection and Training:** Traditional ML classifiers, deep learning architectures, and transformer-based models
5. **Evaluation and Validation:** Train/Test split or cross-validation, independent hold-out testing, comprehensive metrics

6. **Performance Comparison:** Hyperparameter tuning, cross-dataset generalization testing
7. **Deployment and Monitoring:** Model deployment and performance monitoring

2.6 Machine Learning Classifiers

The following machine learning classifiers were evaluated: Support Vector Machine (SVM), Logistic Regression, Decision Tree Classifier, Gradient Boosting, and Random Forest algorithms. These classifiers provide a comprehensive comparison of traditional machine learning approaches for fake news detection.

2.7 Deep Learning Models

Deep learning architectures evaluated include LSTM Networks, Bidirectional LSTM, Convolutional Neural Networks (CNN), and the Bi-LSTM-GRU-Dense Ensemble. These models capture sequential patterns and long-term dependencies in news text.

2.8 Transformer-Based Models

Pre-trained models evaluated include BERT (Base), RoBERTa, and DistilBERT. These transformer-based approaches learn context-aware representations from massive unlabeled corpora, providing state-of-the-art performance in language understanding tasks.

3 Results

3.1 Machine Learning Classifier Performance

Table 1 presents comparative evaluation of five machine learning classifiers on the Indonesian Hoax Dataset (4,231 training articles, 1,058 test articles):

Table 1: Machine Learning Classifier Performance (Indonesian Dataset)

Classifier	Accuracy	Sensitivity	Specificity	AUC
Support Vector Machine (SVM)	83.55%	83.72%	78.12%	0.7161
Logistic Regression (LR)	82.61%	82.73%	75.00%	0.7526
Decision Tree (DTC)	75.33%	83.57%	28.03%	0.5489
Gradient Boosting (GBC)	82.61%	82.85%	70.00%	0.6944
Random Forest (RF)	82.61%	82.85%	70.00%	0.6944

The SVM classifier demonstrated superior performance with 83.55% accuracy, though Logistic Regression achieved the highest AUC (0.7526), indicating better discrimination capability at varying classification thresholds. Decision Tree exhibited high sensitivity (83.57%) but significantly lower specificity (28.03%), indicating propensity toward false positives.

3.2 Ensemble Methods Performance

Table 2 presents ensemble method performance on the Urdu language dataset (4,097 articles across nine domains):

Table 2: Ensemble Methods vs. Individual Classifiers (Urdu Dataset)

Model	Accuracy	Sensitivity	Specificity	MCC	F1 Score
Random Forest (RF)	91.05%	97.15%	81.94%	81.51	89.36%
Extra Trees (ET)	90.81%	97.15%	81.38%	81.02	88.85%
Logistic Regression (LR)	89.10%	93.08%	83.16%	77.18	87.80%
K-Nearest Neighbors (KNN)	78.45%	85.07%	80.17%	54.62	77.20%
Support Vector Machine (SVM)	89.59%	94.71%	81.95%	78.26	88.96%
Stacking Ensemble	93.82%	94.49%	93.50%	86.10	93.17%

The stacking ensemble approach, utilizing Random Forest and Extra Trees as base learners with Logistic Regression as meta-learner, achieved 93.82% accuracy and 86.10 MCC, substantially exceeding any individual classifier. The ensemble method demonstrated balanced sensitivity (94.49%) and specificity (93.50%), indicating superior generalization.

3.3 Deep Learning Model Performance

Table 3 presents deep learning approaches on the LIAR dataset (12.8K political statements):

Table 3: Deep Learning Models on LIAR Dataset

Architecture	Dataset	Accuracy	F1 Score	Notes
CNN	LIAR	27.0%	—	Superior to RNN variants
Bi-LSTM	LIAR	20.4%	—	Prone to overfitting
Bi-LSTM-GRU-Dense Ensemble	LIAR	89.8%	91.4%	Best DL performance
GRU	LIAR	21.7%	—	Moderate performance
CapsNet	LIAR + Features	64.9%	—	With semantic features

The Bi-LSTM-GRU-Dense ensemble model achieved 89.8% accuracy on LIAR dataset, demonstrating the effectiveness of combining multiple RNN variants with dense layers. Integration of metadata features substantially improved classification accuracy.

3.4 Transformer-Based Models Performance

Table 4 summarizes pre-trained language model performance:

Table 4: Transformer-Based Models Performance Summary

Model	Accuracy	AUC	ROC Curve	Advantages
BERT (Base)	85–90%	0.90–0.95	Superior	Context-aware
RoBERTa	87–92%	0.92–0.96	Excellent	Optimized training
DistilBERT	84–89%	0.89–0.94	Good	40% smaller, 60% faster
Fine-tuned Models	91–93%	0.93–0.97	Excellent	Task-specific

Transformer models demonstrated superior contextual understanding, particularly when fine-tuned on fake news-specific tasks. The multimodal BERT-CNN-BiLSTM architecture achieved 81.4% multimodal discrimination accuracy.

3.5 Cross-Validation Results

Table 5 presents five-fold cross-validation results on the Urdu dataset:

Table 5: 5-Fold Cross-Validation Results (Urdu Dataset)

Model	Avg. Accuracy	Std. Dev.	MCC	Robustness
Random Forest	90.38%	$\pm 1.2\%$	80.08	Good
Extra Trees	91.09%	$\pm 0.9\%$	81.75	Good
SVM	91.43%	$\pm 1.8\%$	80.82	Fair
Logistic Regression	89.21%	$\pm 2.1\%$	77.45	Fair
Stacking Ensemble	92.68%	$\pm 0.6\%$	84.76	Excellent

The stacking ensemble exhibited lowest standard deviation ($\pm 0.6\%$), indicating superior consistency across different data splits and enhanced generalization capabilities.

4 Discussion

4.1 Key Findings

The comprehensive investigation reveals several critical insights:

4.1.1 Superiority of Ensemble Methods

Across diverse datasets (Indonesian, Urdu, English), ensemble approaches, particularly stacking, consistently outperformed individual classifiers. The stacking ensemble achieved 93.82% accuracy on the Urdu dataset, representing 5–9 percentage points improvement over the best individual classifier. This superiority stems from ensemble methods' ability to capture complementary patterns recognized by diverse base learners.

4.1.2 Trade-offs Between Feature Types

Traditional feature engineering (TF-IDF, n-grams) provides interpretability and computational efficiency, while transformer-based contextual embeddings capture deeper semantic patterns at substantial computational cost. The Indonesian study demonstrated that SVM with TF-IDF features achieved 83.55% accuracy comparable to more computationally expensive approaches. However, transformer models showed superior performance on shorter text and multimodal tasks.

4.1.3 Language and Domain Specificity

Performance metrics varied substantially across linguistic contexts and domains. The Urdu study achieved 93.82% accuracy across nine domains, while the Indonesian study achieved 83.55% on mixed-domain data. This variation suggests that language-specific preprocessing and domain-adaptive models are essential for robust cross-lingual performance.

4.1.4 Metadata Integration Importance

Incorporating metadata features (speaker profile, publication venue, temporal information) substantially improved classification performance. The LIAR dataset studies demonstrated that metadata features provided discriminative signals complementary to content-based features.

4.1.5 Overfitting in Deep Learning

While deep learning approaches demonstrated impressive performance, susceptibility to overfitting was noted, particularly with limited training data. Ensemble regularization and proper cross-validation mitigated this issue.

4.2 Practical Implications

For production systems, the choice between ensemble, traditional ML, and DL approaches depends on specific constraints:

- **Resource-Constrained Environments:** Logistic Regression with TF-IDF features provides 89.1% accuracy with minimal memory footprint
- **High-Accuracy Requirements:** Stacking ensembles or fine-tuned transformer models achieve optimal accuracy (93–94%)
- **Real-time Applications:** Smaller models (DistilBERT, SVM) enable deployment in latency-sensitive environments
- **Explainability Requirements:** Tree-based models and attention mechanisms provide superior interpretability

5 Future Work and Conclusions

5.1 Future Research Directions

5.1.1 Explainable AI for Fake News Detection

Develop comprehensive explainability mechanisms integrating LIME, SHAP, and attention visualization techniques. Create interpretable frameworks enabling stakeholders to understand classification decisions and build trust in automated systems.

5.1.2 Multimodal Deep Learning Frameworks

Advance multimodal approaches integrating text, images, video, audio, and user behavioral data through sophisticated fusion mechanisms. Investigate how visual manipulation detection complements textual analysis.

5.1.3 Multilingual and Low-Resource Language Support

Extend research beyond English, Indonesian, and Urdu to additional low-resource languages. Develop multilingual pre-trained models and transfer learning approaches enabling effective detection in under-resourced linguistic contexts.

5.1.4 Adversarial Robustness

Systematically evaluate fake news detectors against adversarial attacks. Develop robust training procedures improving model resilience against sophisticated evasion attacks.

5.1.5 Temporal and Propagation Pattern Analysis

Incorporate information diffusion dynamics, temporal evolution, and social network propagation patterns. Develop models capturing how misinformation spreads differently than verified information.

5.1.6 Domain Adaptation and Transfer Learning

Investigate techniques enabling models trained on political misinformation to generalize to health, science, or financial domains. Develop domain-adaptive architectures reducing annotation requirements for new domains.

5.1.7 User and Source Credibility Integration

Develop frameworks incorporating user profile analysis, source reputation tracking, and historical accuracy records into classification pipelines.

5.1.8 Real-Time Deployment and Scalability

Develop production-grade systems capable of processing information streams in real-time. Investigate edge computing approaches and distributed architectures enabling global-scale deployment.

5.2 Conclusions

This comprehensive investigation of fake news detection methodologies across multiple languages, datasets, and approaches reveals that automated detection is achievable with high accuracy (91–94%) through careful combination of feature engineering, algorithm selection, and ensemble techniques. Key conclusions include:

1. **Methodological Convergence:** Ensemble methods effectively combine complementary strengths of machine learning, deep learning, and transformer-based approaches. The optimal approach depends on specific deployment constraints.
2. **Feature Engineering Remains Critical:** Despite deep learning’s automatic feature extraction capabilities, carefully engineered linguistic, metadata, and content-based features substantially improve classification performance.
3. **Multilingual Research Progress:** The development of high-accuracy models for Indonesian (83.55%, SVM), Urdu (93.82%, stacking ensemble), and English datasets represents important progress toward global fake news detection.
4. **Persistence of Fundamental Challenges:** Despite technical achievements, critical challenges remain unresolved: model interpretability, generalization across domains and languages, robustness against adversarial attacks, and integration of multimodal information.
5. **Imperative for Responsible Deployment:** High technical accuracy is necessary but insufficient for responsible deployment. Systems must provide explanations, acknowledge

uncertainty, integrate with human-in-the-loop verification, and address fairness, bias, and transparency concerns.

The proliferation of fake news represents a societal challenge requiring sustained technical innovation complemented by policy interventions, media literacy programs, and collaborative fact-checking infrastructure. Automated detection systems, while valuable tools, constitute one component within comprehensive information integrity ecosystems.

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